

Problem Set 3

Your name

Your Collaborators

Total Number of Points: 40.

Instructions: Solutions should be typeset in $\text{\LaTeX}$ and submitted as PDF by 11:59:59 PM ET on the due date. You can collaborate on the psets in small groups (of up to three), but must write your solutions separately, in your own words. This means not copying from a board nor from your collaborator's computer.

We strongly recommend **not** using AI tools to solve your problem sets. You are responsible for understanding everything you submit and you must be able to explain and justify every part of your solution. In particular, the oral problem set review (10% of the course grade) will evaluate your understanding of the solutions you submit. In all cases, you must list your collaborators, human or machine.

Problem 1. Authenticated Encryption. (12 points) Suppose Alice and Bob want to transmit messages to each other in a *private* and *authenticated* manner.

Let $(\text{Gen}_{\text{Enc}}, \text{Enc}, \text{Dec})$ be an IND-CPA secure symmetric encryption scheme, and let $(\text{Gen}_{\text{MAC}}, \text{Mac}, \text{Ver})$ be an EUF-CMA secure message authentication scheme (that is, existentially unforgeable under a chosen message attack). *You may assume in this problem that $(\text{Gen}_{\text{Enc}}, \text{Enc}, \text{Dec})$ has perfect correctness.*

Alice and Bob share the keys $k_1 \leftarrow \text{Gen}_{\text{Enc}}$ and $k_2 \leftarrow \text{Gen}_{\text{MAC}}$, and now they want to transmit a message M . For each of the following protocols, determine whether it is (i) IND-CPA secure and (ii) EUF-CMA secure. For each of (i) and (ii), if the protocol is secure, give a proof of its security, and if the protocol is not secure, give a counterexample that breaks its security.

- (a) (4 points) $\text{Enc}_{k_1}(M \parallel \text{Mac}_{k_2}(M))$
- (b) (4 points) $\text{Enc}_{k_1}(M) \parallel \text{Mac}_{k_2}(M)$
- (c) (4 points) $\text{Enc}_{k_1}(M) \parallel \text{Mac}_{k_2}(\text{Enc}_{k_1}(M))$

Problem 2. Average case to worst case DDH. (12 points) Let $\mathbb{G}$ be a cyclic group of prime order q generated by $g \in \mathbb{G}$. Let $n = \text{len}(q)$, be the number of bits required to represent q . Assume that we have an efficient algorithm $\mathcal{A}$ that can distinguish a random triple from a non-triple with non-negligible advantage, that is, for $\alpha, \beta, \gamma \leftarrow \mathbb{Z}_q, g \leftarrow \mathbb{G}$ sampled uniformly at random, we have

$$|\Pr[\mathcal{A}(g, g^\alpha, g^\beta, g^\gamma) = 1] - \Pr[\mathcal{A}(g, g^\alpha, g^\beta, g^{\alpha\beta}) = 1]| \geq \frac{1}{p(n)} \quad (1)$$

for some polynomial $p(n)$. Note that this algorithm requires α, β, γ, g to be sampled at random.

- (a) (9 points) Given $\mathcal{A}$, construct an efficient algorithm $\mathcal{B}$ that can distinguish triples for all inputs with negligible error, that is, $\forall \alpha, \beta, \gamma \in \mathbb{Z}_q$ and $\forall g \in \mathbb{G}$ such that $\gamma \neq \alpha\beta$ and g is not the identity, we have

$$|\Pr[\mathcal{B}(g, g^\alpha, g^\beta, g^\gamma) = 1] - \Pr[\mathcal{B}(g, g^\alpha, g^\beta, g^{\alpha\beta}) = 1]| \geq 1 - \mu(n), \quad (2)$$

for some negligible function $\mu(n)$.

- (b) (3 points) Given $\mathcal{A}$, construct an efficient algorithm $\mathcal{B}'$ that can distinguish, with non-negligible probability, (g, g^α, g^β) from $(g, g^\alpha, g^{\alpha^2})$ for all $\alpha, \beta \in \mathbb{Z}_q$ and generator $g \in \mathbb{G} \setminus \{1\}$, given $\beta \neq \alpha^2$.

Problem 3. Circular Security. (8 points) In this problem, we consider a security property (of encryption schemes) called circular security: namely, security even when an adversary is given an encryption of the secret key sk .

One variant of circular security for secret key encryption schemes is defined as follows:

Definition 1 (Circularly Secure SKE) A secret key encryption scheme $(\text{Gen}, \text{Enc}, \text{Dec})$ is said to be circularly secure if for all p.p.t. oracle algorithms $\mathcal{A}^{\mathcal{O}(\cdot)}$, we have that

$$\left| \Pr_{sk \leftarrow_R \text{Gen}(1^n)} [\mathcal{A}^{\text{Left}_{sk}(\cdot)}(c^*) = 1] - \Pr_{sk \leftarrow_R \text{Gen}(1^n)} [\mathcal{A}^{\text{Right}_{sk}(\cdot)}(c^*) = 1] \right| \leq \text{negl}(n), \quad (3)$$

where $c^* \leftarrow \text{Enc}(sk, sk)$, $\text{Left}_{sk}(\cdot)$ is an oracle that on input (m_L, m_R) outputs an encryption of m_L , and $\text{Right}_{sk}(\cdot)$ is an oracle that on input (m_L, m_R) outputs an encryption of m_R .

Think about why this is a reasonable definition and how it is similar to the IND-CPA security discussed in lectures.

An analogous definition for public key encryption is as follows.

Definition 2 (Circularly secure PKE) A public key encryption scheme $(\text{Gen}, \text{Enc}, \text{Dec})$ is said to be circularly secure if any p.p.t. algorithm $\mathcal{A}$ wins the following game (interacting with a challenger) with probability at most $\frac{1}{2} + \text{negl}(n)$:

- (i.) The challenger samples $(pk, sk) \leftarrow \text{Gen}(1^n)$ and sends (pk, c^*) to $\mathcal{A}$, where $c^* \leftarrow \text{Enc}(pk, sk)$ is a ciphertext where the message is the secret key.
- (ii.) $\mathcal{A}$ sends two messages (m_0, m_1) to the challenger.
- (iii.) The challenger selects $b \leftarrow_R \{0, 1\}$ and sends $c_b \leftarrow \text{Enc}(pk, m_b)$ to $\mathcal{A}$.
- (iv.) $\mathcal{A}$ outputs a bit $\tilde{b}$. We say that $\mathcal{A}$ wins if $\tilde{b} = b$.

It turns out that not every IND-CPA secure public key encryption scheme is also circularly secure. Construct a public-key encryption scheme which is IND-CPA secure but not circularly secure, relying only on the existence of public-key encryption schemes. Prove that your scheme is IND-CPA secure but not circularly secure.

Problem 4. Upgrading Lamport Signatures. (8 points)

Recall Lamport's signature scheme from class, based on a OWF $f : \{0, 1\}^n \rightarrow \{0, 1\}^\ell$, that produces an n^2 -bit signature for an n -bit message:

$\text{Gen}(1^n)$

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1 :  $x_{1,0}, \dots, x_{n,0} \leftarrow \{0, 1\}^n$ 
2 :  $x_{1,1}, \dots, x_{n,1} \leftarrow \{0, 1\}^n$ 
3 :  $sk := (x_{1,0}, \dots, x_{n,0}, x_{1,1}, \dots, x_{n,1})$ 
4 :  $vk := (y_{1,0}, \dots, y_{n,0}, y_{1,1}, \dots, y_{n,1})$ , where  $y_{i,c} = f(x_{i,c})$ 
5 : return  $(sk, vk)$ 
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$\text{Sign}(sk, m \in \{0, 1\}^n)$

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1 : parse  $sk = (x_{1,0}, \dots, x_{n,0}, x_{1,1}, \dots, x_{n,1})$ 
2 : return  $\sigma := (x_{1,m_1}, \dots, x_{n,m_n})$ 
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$\text{Ver}(vk, m \in \{0, 1\}^n, \sigma)$

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1 : parse  $\sigma := (\sigma_1, \dots, \sigma_n)$ 
2 : if  $\forall i \in [n], f(\sigma_i) \stackrel{?}{=} y_{i,m_i}$  : return 1
3 : else : return 0
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In this problem, we will look at a stronger definition of one-time unforgeability known as *one-time strong unforgeability* which states that not only is the adversary unable to produce a signature on a different message, but also that she is unable to produce a *different* signature σ^* on the same message it requested a signature on.

Definition 3 (One-time strong unforgeability)

Let $(\text{Gen}, \text{Sign}, \text{Ver})$ be a digital signature scheme with message space $\mathcal{M}$ and key space $\mathcal{K}$ with security parameter n . This scheme is one-time strongly unforgeable if for all pair of PPT algorithms $(\mathcal{A}_1, \mathcal{A}_2)$, there exists a negligible function negl such that for all n ,

$$\Pr \left[\begin{array}{l} (sk, vk) \leftarrow \text{Gen}(1^n); \\ (m, \text{state}) \leftarrow \mathcal{A}_1(vk); \\ \sigma \leftarrow \text{Sign}(sk, m); \\ (m^*, \sigma^*) \leftarrow \mathcal{A}_2(\sigma, \text{state}) \end{array} : \begin{array}{l} (m^*, \sigma^*) \neq (m, \sigma) \wedge \\ \text{Ver}(vk, m^*, \sigma^*) = 1 \end{array} \right] \leq \text{negl}(n).$$

- (a) (3 points) Show an attack on the one-time strong unforgeability of Lamport's scheme. That is, assuming the existence of length-preserving one-way functions, construct a one-way function f such that the Lamport signature scheme using f is *not* one-time strongly unforgeable.
- (b) (2 points) Prove that if f is an injective one-way function, then Lamport's scheme is one-time strongly unforgeable.
- (c) (3 points) Show that if one-way functions exist, then there exists a one-way function such that Lamport's scheme is one-time strongly unforgeable. Note that building injective one-way functions from arbitrary one-way functions is an open question. (You may use statements from class or earlier in this problem set.)